Serial Decode for the Keithley DMM6500 — notes for beta testers

version 1.72 · `Serial_Decode.tspa` · a TSP app that turns a DMM6500 into a UART/LIN decoder

It digitizes a serial line, finds the bit rate and framing itself, and shows the decoded bytes on the front panel. No host computer is needed once it is loaded.

Everything below is either a documented behaviour that **looks** like a bug, or a defect I already know about. Please read it before reporting anything — it will save you rediscovering my own scars.

Loading it

Copy `Serial_Decode.tspa` to a USB key, insert it, and load it from **Menu** → **Scripts**. Run the script; it builds its own screens.

Relaunch with *End App*, not by reloading from the host. The firmware never reclaims display object IDs, so there is a hard limit on how many times the UI can be built between power cycles — about **4** host reloads, but about **12** if you End App and relaunch the script. When it runs out the screen simply stops being drawn, which looks exactly like a broken script. A power cycle resets it.

Four behaviours that are correct, or known, and not worth reporting

A signal whose voltage band straddles ground is taken to be RS-232 unless the bytes say otherwise. A low level below −1 V with a high level above +1 V reads as RS-232 at line levels, which marks at its *negative* level. On a capture holding an idle interval the resting level is observed and the levels get no say; on one that never goes quiet, both polarities are decoded and the one the bytes support is kept. The case it cannot settle is a line that never idles and carries no text, where the two readings can be exactly equal: there the RS-232 reading stands and polarity stays on auto rather than locking a coin flip. If a DC-shifted single-supply waveform comes back with the right bit rate, the right framing, no errors and every byte wrong, that is this, and `Options ▸ Polarity` overrides it. Say what the two levels were and whether the line ever went idle.

A bit rate printed with a trailing `?` — e.g. `38400?` — means the app snapped the measurement to a standard rate but does not stand behind it. It locks to the rate it *measured*, not the one displayed. That is a deliberate hedge, not a rendering fault.

37500 Bd can read as 38400, and 125000 as 128000. The bit-time fit lands slightly high and the rounding step then takes it to the standard rate next door. The bytes are unaffected, and it is known and unfixed on purpose. **It does not always carry the `?`** — the mark measures the distance from the fit to the standard rate, so a fit that lands close to the wrong one prints a bare number. A reading at a *multiple* of the true rate is a different mechanism; it survives only on impaired signals, so one on a clean line is worth reporting.

7E1 traffic is read as 8N1. The parity bit lands in bit 7, so every byte with odd parity comes back `+0x80` and roughly half the payload looks corrupt. Known.

Logging

The app writes logs to `/usb1/SERDEC`. With no key inserted, or a full one, logging silently does nothing — if you expected a file and got none, check the key first. Capture and decode still work in every mode without a key; the recordings put their bytes on the panel and say `NOT recorded`.

The key can be pulled and replaced while the app runs. The slot is polled twice a second while idle, so the buttons and the log cell follow it within half a second, and a replacement key gets a new name.

Screenshots are better than photographs

The DMM6500 will hand you a pixel-exact 800×480 capture of its own screen over its LXI web interface, with nothing loaded and no script running. It is undocumented by Keithley — I read it out of the instrument's own virtual-front-panel JavaScript:

```
POST http://<ip>/ajax_proc      function=9              -> session=<id>
GET  http://<ip>/images/fp.tga?<session>:<milliseconds>  -> uncompressed TGA, 800x480
GET  http://<ip>/images/fplow.tga?<session>:<ms>         -> half-res 400x240
```

HTTP basic auth, factory default `admin` / `admin`. `tools/screenshot.py` in the repository does this and writes a PNG; change `HOST` to your instrument's address.

Please send the whole frame rather than a crop. The numbers that diagnose most problems live in the header and footer lines, and a screenshot shows what the app *drew* — which is not always what it computed. Several of this project's bugs lived exactly in that gap.

Reporting something

Please open an issue:

https://github.com/ameline-design/dmm6500-serial-decode/issues

Include, as far as you can:

1. **A screen grab** — the whole 800×480 frame, per the section above. If the panel changed as you watched, one before and one after is worth more than either alone.
2. **The text that should have been decoded.** Paste the exact bytes or string you were transmitting. It turns "the decode looks wrong" into a difference I can compute, and it is the one piece I cannot reconstruct from anything else.
3. **The bit rate you set**, as a number, and whether it is a standard rate or something unusual.
4. **The framing you sent** — data bits, parity, stop bits, and the idle polarity if you know it (8N1, 7E1, 8E2 …). If the app reported a framing different from the one you sent, say both.
5. **The signal levels** — low and high in volts, and whether the band crosses ground. Read the first behaviour above before deciding this does not matter; it is the most common cause of a decode where every byte is wrong and every other number is right.
6. **Anything that needed a power cycle** to recover, and what you had just done.

A rough report is far better than none — send what you have and I will ask.

A wrong number that looks plausible is worth more to me than a crash. Crashes I can find on my own; confidently-wrong answers are what this project exists to hunt. I have measured this from the inside for months, so the gaps are unlikely to be in the arithmetic — they are much more likely to be in whether any of

it makes sense to somebody who did not build it. A panel number you cannot interpret, or two that seem to disagree, is a real finding, not a naive question.